

Testing ML Link Adaptation Under RF Impairments: Failure Detection, Diagnosis and Safe Fallback

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Abstract—Machine-learning (ML) features are entering the 5G/6G air interface, but a model that improves throughput on the channels it was trained on can fail silently elsewhere. For test and measurement, the open question is less “is the ML better?” than “where can it be trusted, and how is failure detected?”. We build a simulation framework in the spirit of a software call box: an NR-like PDSCH link with TS 38.101-4 fading profiles and four switchable RF impairments (PA compression, IQ imbalance, phase noise, residual CFO), a BLER-table abstraction for fast closed-loop runs, an outer-loop link adaptation (OLLA) baseline, and an ML link-adaptation model as the device under test. Across 52 channel/impairment scenarios, the ML model beats a tuned OLLA by +2 to +15% on clean fading channels, including one it never saw, but loses up to 89% at high SNR under moderate or severe impairments. A self-check that compares the model’s predicted ACK probability with observed ACKs detects these failures far better than input-distribution monitoring (AUROC 0.89 vs. 0.68). A four-view CNN names the impairment correctly in all 23 high-SNR failure cases, and a fallback to OLLA reduces the average loss in failure cases from -35.8% to -6.9% while keeping most of the gain elsewhere ($+9.0\%$ to $+8.1\%$). One failure remains invisible to the self-check: an over-cautious model on a non-fading channel.

I. INTRODUCTION

Link adaptation (LA) selects a modulation and coding scheme (MCS) for every slot from imperfect channel-state reports. The standard approach, OLLA [1], adds an offset to the reported SNR that moves up on ACK and down on NACK, holding the block error rate (BLER) near a target of 10%. ML-based LA can exploit structure OLLA ignores, but it raises a verification problem that is central to AI/ML in the NR air interface [2]: a model can perform well in its training conditions and degrade outside them without any visible sign.

This work treats that as a test problem and builds a miniature version of the workflow a call-box team would need: *feature* (an ML LA model), *test* (sweeps over conformance channels and RF impairments), *diagnose* (detect and name the cause of failure), and *mitigate* (fall back safely). Our contributions are:

- an open simulation framework with a full NR-like PHY, conformance fading profiles, four calibrated impairment models, and a fast BLER-table closed loop (Sec. II);
- a map of where an ML LA model wins and fails across 52 scenarios (Sec. IV-B);
- a comparison of two failure detectors using only gNB-side information, showing that a prediction self-check

outperforms input out-of-distribution (OOD) detection (Sec. IV-C);

- an impairment classifier on instrument-style views, and a diagnosis-aware fallback that removes most of the downside of the ML model (Secs. IV-D, IV-E).

II. SYSTEM MODEL

A. Link-level chain

We simulate a single-layer downlink with Sionna [3] (PyTorch backend): 30 kHz subcarrier spacing, 24 PRBs, FFT size 512 ($f_s = 15.36$ MHz), normal cyclic prefix, 14-symbol slots, and type-1 comb DMRS on symbols 2 and 11 with no data on DMRS symbols, leaving 3456 data resource elements per slot. MCS and transport-block size follow TS 38.214 Table 5.1.3.1-1 [4]; MCS 0–2 are excluded because the LDPC encoder used does not support rates below 1/5. The receiver uses least-squares channel estimation with linear interpolation in frequency and time, MRC, soft demapping, and 20-iteration belief-propagation LDPC decoding.

Channels are the TS 38.101-4 [5] conformance profiles TDLA30-10, TDLB100-400 and TDL300-100 (TR 38.901 TDL models [6]) plus AWGN. The channel is held constant within each OFDM symbol, which captures channel ageing between DMRS symbols but ignores Doppler-induced ICI (below 1.5% of the subcarrier spacing here).

B. RF impairments

Impairments are placed where they occur in a downlink test: PA compression at the transmitter (call-box side), and residual CFO, phase noise and IQ imbalance at the UE receiver, applied after thermal noise. The PA follows the Rapp model [7] with smoothness $p = 2$ and input back-off as the severity parameter. IQ imbalance is $y = \mu x + \nu x^*$ with gain and phase mismatch; phase noise is a Wiener process with a given 3-dB linewidth; CFO is a constant rotation normalised to the subcarrier spacing. Each has mild, moderate and severe settings calibrated to cap post-equalisation SINR at roughly 27–30, 18–21 and 12–16 dB (AWGN, 45 dB SNR). With the clean case this gives 13 settings, and with the four channels, 52 scenarios.

C. BLER tables and closed loop

Decoding is too slow for long closed-loop runs, so each scenario is simulated once (on a GPU) and summarised per MCS by a three-parameter fit against the *slot SNR*, the wideband SNR of that slot’s fading realisation:

$$\text{BLER}(s) = 1 - (1 - f) \sigma(k(s - s_{50})), \quad (1)$$

with centre s_{50} , steepness k and error floor f , fitted by maximum likelihood to per-block outcomes. The slot SNR excludes impairments: the scheduler cannot see them, so they appear only as shifted curves or error floors. Conditioning on wideband SNR treats in-slot frequency selectivity as randomness inside each curve rather than mapping it exactly, as EESM/MIESM abstractions would.

The closed loop runs one iteration per 0.5 ms slot. The true slot SNR follows a TDL fading process sampled at the slot rate. The scheduler sees an SNR report every 5 ms, measured 2 ms earlier, with 1 dB error and 1 dB quantisation, plus noisy Doppler and delay-spread estimates, and receives ACK/NACK after 1 ms. ACKs are drawn from the BLER table at the true slot SNR of the true scenario. There are no HARQ retransmissions, and a single full-buffer user is assumed.

III. LINK ADAPTATION UNDER TEST

A. Baseline: OLLA

The inner loop picks the highest MCS whose AWGN-curve SNR for 10% BLER is below the report plus offset; the offset decreases by $\Delta_{\downarrow}$ on NACK and increases by $\Delta_{\uparrow} = \Delta_{\downarrow} t / (1 - t)$ on ACK, so the BLER converges to $t = 0.1$. A sweep of $\Delta_{\downarrow} \in [0.1, 2]$ dB showed throughput within 1–4% across step sizes; we use 0.5 dB. OLLA holds 10.0% BLER in all four clean channels.

B. Device under test: ML link adaptation

The model sees ten gNB-side features: the latest report, its trend and age, log Doppler and delay-spread estimates, ACK rates over the last 8 and 32 feedbacks, a background OLLA-style offset, the last MCS, and the current NACK streak. An MLP (two hidden layers of 128 units) outputs, for every MCS at once, the probability that a block sent with that MCS will be ACKed. Because the BLER tables give a label for every MCS in every slot, training needs no exploration: targets are $1 - \text{BLER}_m$ at the true slot SNR, with binary cross-entropy loss. The MCS is then chosen either to maximise $\text{TBS} \times P(\text{ACK})$ or, in the mode we evaluate unless stated, as the highest MCS with $P(\text{ACK}) \geq 0.9$ (same target as OLLA).

Training data come from two rounds: closed-loop runs with an exploring OLLA, then runs with the round-0 model itself (Dagger [8]), so the model also sees the ACK patterns its own decisions produce. Training is deliberately limited to TDLA30-10 and TDLB100-400 without impairments, at average SNRs drawn from 0–28 dB. Everything else is out of distribution.

TABLE I

E3: DETECTORS OVER 3744 WINDOWS OF 250 MS (22.8% FAILURES)

Detector	AUROC	Caught	False alarms
Input OOD (Mahalanobis)	0.68	66%	39%
Prediction self-check	0.89	66%	20%

False alarms: share of non-failure windows flagged (0% in training conditions for both).

IV. EXPERIMENTS AND RESULTS

A. E1: ML vs. OLLA on clean channels

Fig. 1 compares the ML model with OLLA and with an oracle that knows the true slot SNR (3 seeds, 5 s of traffic each). In the training channels, the ML model gains +2 to +9% (TDLA30-10) and +7 to +13% (TDLB100-400) at about the same BLER. It generalises to the unseen TDL300-100 (+9 to +15%), but on the unseen AWGN channel it loses 21% at 5–10 dB: its inputs (no Doppler, no delay spread) lie outside the training range and it becomes over-cautious (0% BLER). One mechanism behind the gain is visible in the tables: under fading, MCS 16 needs more SNR than MCS 17 while carrying less data. OLLA’s AWGN-based mapping still uses MCS 16 (about 12% of slots at 13 dB in TDLA30-10); the ML model never does.

B. E2: where the ML model fails

We ran the ML model and OLLA on identical fading and noise (common random numbers) for all 52 scenarios at 5, 15 and 25 dB (2 seeds, 3 s each). The ML model is more than 5% worse than OLLA in 34 of 156 cells, with a worst case of –89%. Fig. 2 (left) shows the pattern: failures concentrate at 25 dB with moderate or severe impairments. There, the reports promise a high SNR, but the impairment caps the achievable SINR below what the chosen MCS needs. At 5 dB the same impairments are buried under thermal noise and the model still gains about 2–8%. OLLA is robust by construction, since its ACK-driven offset absorbs any constant SNR penalty the reports hide.

C. E3: detecting failure

Both detectors use only information a gNB or call box has. *Input OOD* scores how far the feature vector is from the training data (Mahalanobis distance [9], averaged over 250 ms). The *self-check* compares the model’s mean predicted $P(\text{ACK})$ for its chosen MCS with the observed ACK rate in the same window. Thresholds are the 99th percentile over training-condition windows on separate seeds. A window is labelled a failure if ML delivers more than 5% less data than OLLA in it.

Table I shows the self-check is clearly better. Input OOD flags TDL300-100, whose long delay spread is unusual but where the model does well, and misses impairments that leave the inputs unchanged. “The inputs look unusual” is a weak proxy for “the model is wrong.” Neither detector sees the AWGN failure: at 5 dB the model predicts 97% ACK for MCS 5 and observes 100%, so its own choices look fine. The

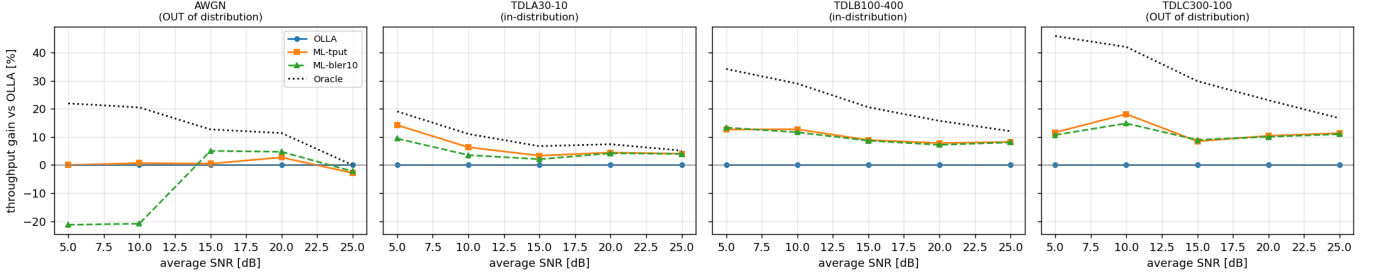


Fig. 1. E1: throughput gain over OLLA on clean channels. TDLA30-10 and TDLB100-400 are training conditions; AWGN and TDL300-100 were never seen. The oracle knows the true slot SNR and is an upper bound.

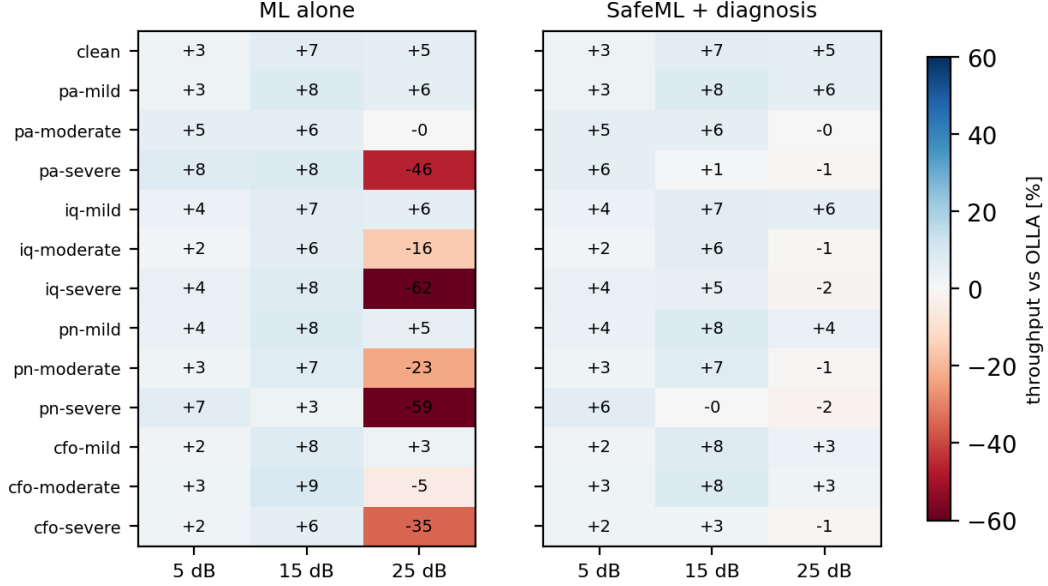


Fig. 2. E2/E5: throughput vs. OLLA per impairment setting and SNR, averaged over the four channels. Left: ML alone. Right: ML with the self-check, fallback and diagnosis.

loss lies in the higher MCS it never tries, and detecting it requires probing, which OLLA does by design.

D. E4: naming the impairment

A small CNN classifies one slot of equalised symbols into clean, PA, IQ, phase noise or CFO. Instead of the constellation alone, it receives four instrument-style 32×32 views (Fig. 4): the constellation; radial error vs. amplitude; *image leakage*, the error on subcarrier k multiplied by the decided symbol on $-k$, which IQ imbalance shifts off-centre; and phase error vs. OFDM symbol, where CFO drifts and phase noise wanders. Reference symbols are hard decisions, so no genie data is used. The CNN also gets the SNR, which a call box knows because it sets the noise level.

Trained on 8000 slots across all channels, SNRs of 5–35 dB and all severities, and tested on 2400 unseen slots, the constellation-only CNN reaches 46% and the full model 59% (5 classes, Fig. 3). For moderate and severe impairments at $\text{SNR} \geq 20$ dB, accuracy is about 95%. The image-leakage view cuts IQ-to-PA confusion from 0.41 to 0.17.

PA compression does not appear as an inward pull of the outer points: in OFDM the clipping distortion spreads over all subcarriers and behaves like additive noise (Bussgang [10]), so it is only recognisable as more noise than the SNR explains. Mild impairments, and any impairment at low SNR, are mostly undiagnosable because their distortion is below thermal noise.

E. E5: safe fallback

SafeML runs the ML model while keeping OLLA warm on the same ACKs. Over the last 200 resolved slots it compares predicted and observed ACK rates; if the model is overconfident by more than 0.16, control passes to OLLA. With diagnosis enabled, an impairment verdict keeps OLLA in control (hardware faults persist), while “clean” lets ML retry after 1 s. Diagnosis is only trusted at $\text{SNR} \geq 15$ dB, following E4. A first version with a 0.08 margin was triggered by ordinary slow fades and kept only +2.4% of the ML gain; real failures show gaps of 0.3–0.8, so the wider margin loses nothing in detection.

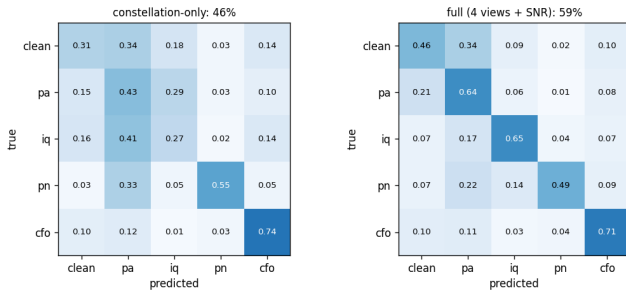


Fig. 3. E4: confusion matrices (constellation-only vs. four views + SNR), accuracy vs. SNR, and accuracy by severity.

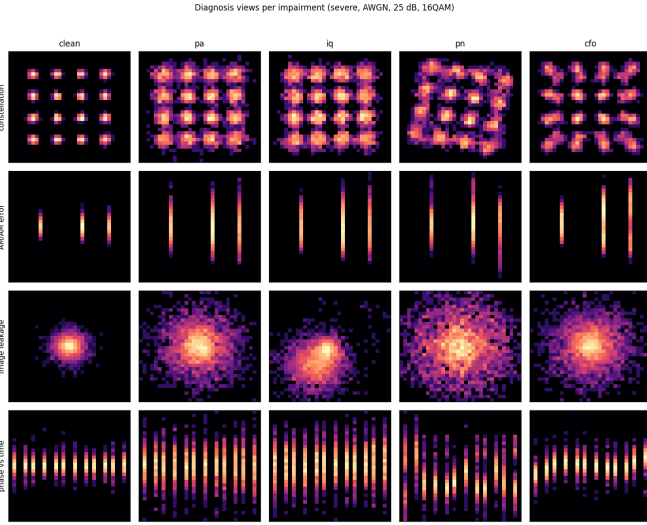
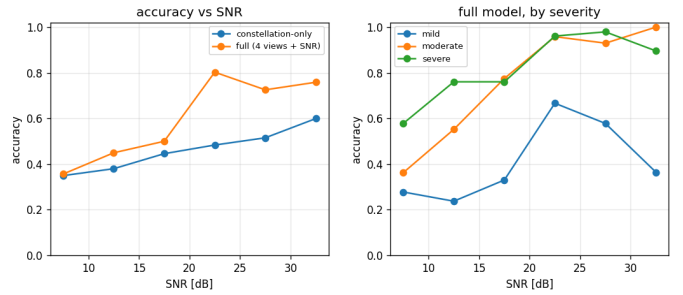


Fig. 4. Diagnosis views for each class (severe level, AWGN, 25 dB, 16QAM).

TABLE II
E5: THROUGHPUT VS. OLLA OVER 156 SCENARIO/SNR CELLS

Policy	ML-fail cells	ML-win cells	Worst
ML alone	-35.8%	+9.0%	-89%
SafeML	-9.1%	+8.2%	-21%
SafeML + diagnosis	-6.9%	+8.1%	-21%

34 cells where ML alone is $> 5\%$ below OLLA; 102 where it is $> 2\%$ above.

Table II and Fig. 5 show the result: the fallback removes almost all of the downside and keeps almost all of the upside. The diagnosis named the correct impairment in all 23 high-SNR failure cells. The remaining worst case (-21%) is AWGN at 5 dB, the over-caution failure no self-check can see.

V. LIMITATIONS

All results are simulation. The impairment models are simplified and not calibrated to specific hardware. The BLER abstraction conditions on wideband SNR and omits HARQ combining. There is no PTRS, so phase noise is uncorrected between DMRS symbols, which is pessimistic for FR1. The diagnosis assumes one impairment at a time. The stress test

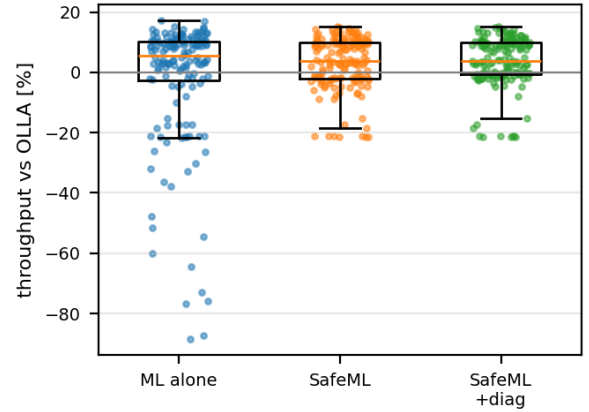


Fig. 5. E5: throughput vs. OLLA in each of the 156 cells, per policy.

uses 2 seeds per cell; E1 uses 3. The harness exposes a mock instrument interface so a real call box could replace the simulator, but this has not been tested.

VI. CONCLUSION

An ML link-adaptation model can beat a tuned OLLA by up to 15% and still fail by up to 89% under impairments its inputs cannot reveal. For testing such features, three findings stand out: (i) failure regions are structured (high SNR, significant impairment) and can be mapped by sweeping conformance channels against impairment severity; (ii) a self-check of the model's own predictions detects failure far better than input-distribution monitoring, but cannot see over-caution; (iii) instrument-style views plus the known SNR make the cause of failure identifiable, enabling a fallback that keeps the ML gain where it is real. Natural next steps are combined impairments, HARQ-aware abstraction, probing to expose over-caution, and validation against measured impairment data.

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